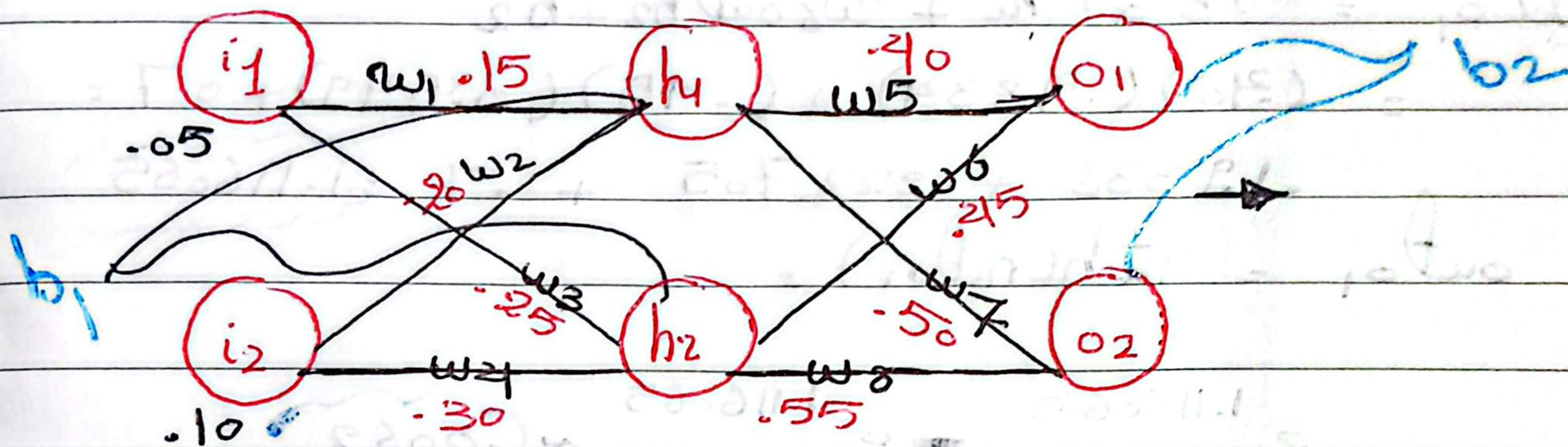


Assignment 1

Forward

write the Python Code for the previous network for the Tanh activation function;
choose the weight random from interval $[-0.5, 0.5]$
$b_1, b_2 = 0.5, 0.7$ respectively.
print the output of the network.



Solution for this problem?

inputs $i_1 = -0.05$ & $i_2 = -1.0$

weights $w_1 = 0.15$ & $w_2 = -0.20$ & $w_3 = 0.25$

& $w_4 = -0.30$ & $w_5 = 0.40$ & $w_6 = 0.25$ & $w_7 = -0.50$

& $w_8 = -0.55$

Biases

$b_1 = 0.5$ & $b_2 = 0.7$

Function

Tanh

$$\text{net } h_1 = w_1 i_1 + w_2 i_2 + b_1$$

$$(.15)(-.05) + (-.20)(-1.0) + .5$$

$$-.0075 + .20 + 0.5 = .5925$$

$$\text{out } h_1 = \tanh(\text{net } h_1) =$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} =$$

$$\frac{e^{.5275} - e^{-.5275}}{e^{.5275} + e^{-.5275}} = \frac{1.6947 - 0.5901}{1.6947 + 0.5901}$$

$$\approx 0.4834$$

$$\text{net } o_1 = w_5 \text{out } h_1 + w_6 \text{out } h_2 + b_2$$

$$= (.20)(.4834) + (.25)(.4949) + 0.7 =$$

$$.19336 + .222705 + .7 = 1.116065$$

$$\text{out } o_1 = \tanh(\text{net } o_1) =$$

$$\frac{e^{1.116065} - e^{-1.116065}}{e^{1.116065} + e^{-1.116065}} \approx 0.8062$$

$$\text{net } o_2 = w_7 \text{out } h_1 + w_8 \text{out } h_2 + b_2$$

$$\text{net } h_2 = w_3 i_1 + w_4 i_2 + b_1$$

$$(-.25)(-.05) + (-.30)(-.10) + 0.5$$

$$.0125 + .03 + .5 = .5425$$

$$\text{out } h_2 = \tanh(.5425) =$$

$$\frac{e^{.5425} - e^{-.5425}}{e^{.5425} + e^{-.5425}} \approx 0.4949$$

$$\text{net } o_2 = w_7 \text{out } h_1 + w_8 \text{out } h_2 + b_2$$

$$net\ o_2 = (.50)(0.4834) + (-.55)(0.4924) + 0.7 =$$

$$.2417 + .272195 + .7 = 1.213895$$

$$out\ o_2 = \tanh(1.213895) =$$

$$\frac{e^{1.213895} - e^{-1.213895}}{e^{1.213895} + e^{-1.213895}} \approx .8379$$

$$\Rightarrow out\ o_1 \approx .8062$$

$$\Rightarrow out\ o_2 \approx .8379$$

Total Error:

$$t_1 = .01$$

$$out_1 \approx .8062$$

$$t_2 = .99$$

$$out_2 \approx .8379$$

$$\text{Squared Error} = \sum \frac{1}{2} (\text{target} - \text{output})^2 =$$

$$E_{o1} = \frac{1}{2} (t_1 - out_{o1})^2 =$$

$$\frac{1}{2} (.01 - .8062)^2 =$$

$$\begin{aligned} & \cdot 31696722 \\ & \cdot 31696722 \end{aligned}$$

$$E_{o2} = \frac{1}{2} (t_2 - out_{o2})^2 =$$

$$\frac{1}{2} (.99 - .8379)^2 = 0.011567205$$

$$\text{Total Error} = E_{o1} + E_{o2} =$$

$$\cdot 31696722 + 0.011567205 \approx \boxed{.3285}$$